Experiment - 1 : Comparative Study of Low-Code Development Versus Traditional Application Development Approaches

Aim:

To compare low-code development and traditional application development approaches based on development speed, ease of use, customization, cost, maintenance, scalability and overall suitability, using the **Recruitment & Job Application Management** application as the case study.

Theory:

Application development is the process of designing, developing, testing and deploying software applications. Two major approaches are followed today.

Low-Code Development: Low-code platforms such as Zoho Creator provide visual builders, drag-and-drop fields, ready-made report types and workflow designers, so that an application can be built with very little hand-written code. Logic that is not available visually is added through a small scripting language (Deluge in Zoho Creator). Low-code suits business process applications such as recruitment, leave management and event registration.

Traditional Application Development: Traditional development uses programming languages such as Java, Python, C# or JavaScript with frameworks and databases. The developer controls every layer (UI, business logic, database, deployment), which gives full flexibility but needs more time, skill and testing.

Case Study - Job Application System: The recruitment application needs a candidate application form, nine types of reports (List, Spreadsheet, Kanban, Calendar, Map, Timeline, Pivot Table, Chart and Summary), email notifications and a dashboard. In Zoho Creator each of these is a built-in component. In a traditional stack the same features need a database schema, REST APIs, a front-end framework, a charting library, a mapping API and an email service.

Comparison:

Parameter	Low-Code Development (Zoho Creator)	Traditional Development
Development Time	Faster - form and nine reports built in hours	Slower - weeks for UI, API and database
Coding Requirement	Minimal - only Deluge for validation and emails	Extensive - full source code
Learning Curve	Easy	Moderate to difficult
Customization	Limited to platform options	Highly flexible
Reports and Charts	Built-in report types and page charts	Must be coded with libraries
Cost	Lower for small and medium projects	Higher initial cost
Scalability	Moderate (plan limits apply)	High
Maintenance	Easier, platform handles hosting and updates	Requires skilled developers
Best Suited For	Business apps such as recruitment tracking	Complex enterprise and consumer products

Procedure:

1. Study the concepts of low-code development and traditional application development.
2. Select Zoho Creator as the low-code platform and review its form builder, report types, workflows and pages.
3. List the features required by the Job Application System (form, nine report types, workflows, email, dashboard).
4. Study how the same features would be developed with a traditional stack (database, back-end, front-end, charting and mapping libraries).
5. Compare both approaches based on development time, coding effort, customization, scalability, cost and maintenance.
6. Record the observations in a comparison table.
7. Draw conclusions on the suitable approach for different application requirements.

Result:

The comparative study shows that low-code development greatly reduces development time and coding effort, which makes it suitable for business applications such as the Job Application System. Traditional development needs more time and programming skill but gives better flexibility, scalability and customization, so it is preferred for large and complex software systems.